

The retina uncouples glycolysis and oxidative phosphorylation via Cori-, Cahill-, and mini-Krebs-cycle

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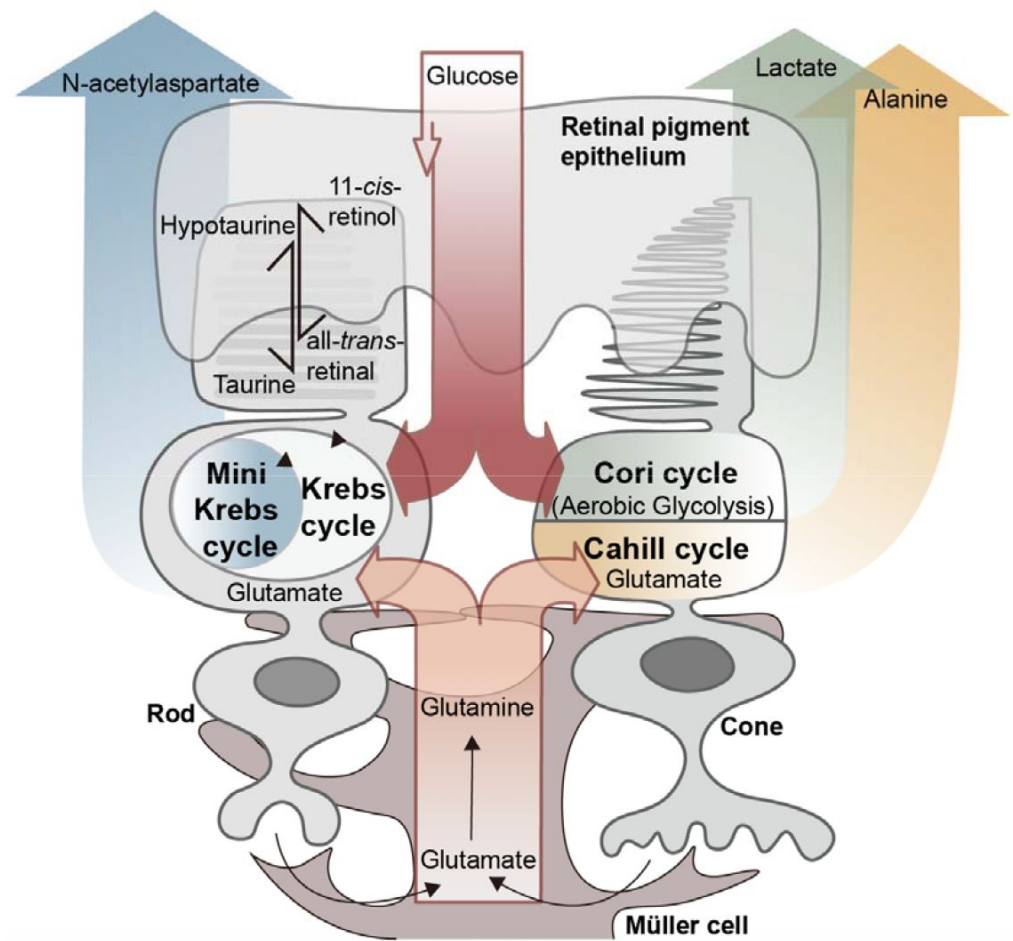
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Abstract

The retina consumes massive amounts of energy, yet its metabolism and substrate exploitation remain poorly understood. Here, we used a murine explant model to manipulate retinal energy metabolism under entirely controlled conditions and utilized ^1H -NMR spectroscopy-based metabolomics, *in situ* enzyme detection, and cell viability readouts to uncover the pathways of retinal energy production. Our experimental manipulations resulted in varying degrees of photoreceptor degeneration, while the inner retina and retinal pigment epithelium were essentially unaffected. This selective vulnerability of photoreceptors suggested very specific adaptations in their energy metabolism. Rod photoreceptors were found to rely strongly on oxidative phosphorylation, but only mildly on glycolysis. Conversely, cone photoreceptors were highly dependent on glycolysis but insensitive to electron transport chain decoupling. Importantly, photoreceptors were able to uncouple glycolytic and Krebs-cycle metabolism via three different pathways: 1) the mini-Krebs-cycle, fueled by glutamine and branched-chain amino acids, generating N-acetylaspartate; 2) the alanine-generating Cahill-cycle; 3) the lactate-releasing Cori-cycle. Moreover, the metabolomic data indicated a shuttling of taurine and hypotaurine between the retinal pigment epithelium and photoreceptors, resulting in an additional net transfer of reducing power to photoreceptors. These findings expand our understanding of retinal physiology and pathology and shed new light on neuronal energy homeostasis and the pathogenesis of neurodegenerative diseases.



Retinal photoreceptors employ both glucose and glutamate as fuels. While rod photoreceptors rely strongly on oxidative phosphorylation and the N-acetylaspartate producing mini-Krebs-cycle, cone photoreceptors rely on the lactate-producing Cori cycle and the oxidative, alanine-producing Cahill cycle.

Highlights

- The retina utilizes a complex energy switchboard consisting of the Krebs cycle, mini-Krebs cycle, Cahill cycle, and Cori cycle.
- Mini-Krebs cycle runs more efficiently than ‘full’ Krebs cycle.
- Alanine transaminase decouples glycolysis from the Krebs cycle.
- Lactate, alanine, and N-acetylaspartate are distinctive energetic pathway signatures.

eLife assessment:

Chen and colleagues utilize an in situ explant model of the neural retina and retinal pigment epithelium (RPE), along with small molecule inhibition of key metabolic enzymes and targeted metabolomic analysis, to decipher key differences in metabolic pathways used by rods, cones, Muller glia, and the RPE. They conclude that rods are heavily reliant on oxidative metabolism, cones are heavily reliant on glycolysis, and multiple mechanisms exist to decouple glycolysis from oxidative metabolism in the retina. This study provides **valuable** metabolomic data and insights into the metabolic flexibility of different retinal cells. However, the evidence supporting the authors' claims is **incomplete** and would benefit from validating their evidence with animal models, comparing their data to previously published literature, performing C13 tracing experiments, and assessing mitochondrial function.

Introduction

The retina is the neuronal tissue with the highest energy demand (1, 2). Paradoxically, the mammalian retina is thought to strongly rely on energy-inefficient glycolysis, even though oxygen is available and high-yield oxidative phosphorylation (OXPHOS) possible (3). This aerobic glycolysis releases large amounts of lactate, as already reported in the 1920s by Otto Warburg (4). High retinal energy demand is linked to the extraordinary single-photon sensitivity of photoreceptors (2, 5, 6), and to lipid synthesis for the constant renewal of photoreceptor outer segments (7, 8).

The retina harbors two types of photoreceptors: rods, which exhibit remarkable light sensitivity and enable night vision; and cones, which work in daylight and allow color vision. Cones spend around twice as much energy as rods (9). Photoreceptors are not connected to the vasculature and are nourished by other retinal cell types, including retinal pigment epithelial (10) cells or Müller glial cells (MGC). Moreover, photoreceptors can experience changes in energy demand on a millisecond timescale, from very high in the dark, to 4-10 times lower in light (5).

Glycolysis provides for rapid but inefficient adenosine triphosphate (ATP) production, while Krebs-cycle and OXPHOS are very efficient but much slower. Since glycolysis and Krebs-cycle are metabolically coupled through pyruvate, it is unclear how photoreceptors adapt to sudden and large changes in energy demand, and what energy substrates they may use. A recent hypothesis proposed that photoreceptors use predominantly aerobic glycolysis, with the resultant lactate utilized by the retinal pigment epithelium (RPE) and MGCs for OXPHOS (11). However, this contrasts with the high density of mitochondria in photoreceptor inner segments. Retinal energy metabolism has obvious ramifications for disease pathogenesis, including diabetic retinopathy, age-related macular degeneration, or inherited retinal diseases, such as retinitis pigmentosa.

Here, we studied expression of energy metabolism-related enzymes in different retinal cell types, using organotypic retinal explants maintained in serum-free, fully defined medium. Metabolic functions of the RPE were investigated by culturing retina with and without RPE. As readouts, we correlated enzyme expression with quantitative proton nuclear magnetic resonance (^1H -NMR) spectroscopy-based metabolomics and cell death. Notably, interventions in energy metabolism caused selective photoreceptor cell death, while leaving other retinal cell types unaffected. Metabolomic analysis and localization of key enzymes identified important pathways and shuttles altered, explaining the strong interdependence of the various retinal cell types, and opening new perspectives for the treatment of neurodegenerative diseases in the retina and beyond.

Retinal expression patterns of key energy metabolism related enzymes

We used immunofluorescence to assess expression and cellular localization of enzymes important for retinal metabolism (**Figure 1**). The outermost retinal layer is formed by RPE cells expressing RPE-specific 65kDa protein (RPE65), dedicated to the recycling of photopigment retinal (12). Hence, RPE65 immunolabeling revealed the RPE cell monolayer (**Figure 1A**).

Immunofluorescence for glucose-transporter-1 (GLUT1) showed strong labelling on basal and apical sides of RPE cells (**Figure 1A**), in line with previous literature (3, 13). Glucose uptake from the RPE into the neuroretina is mediated by high affinity/high capacity GLUT3 (14), strongly expressed on photoreceptor inner segments (Supplementary Figure 1A). Pyruvate kinase is essential to glycolysis, catalyzing the conversion of phosphoenolpyruvate to pyruvate. While pyruvate kinase M1 (PKM1) was expressed in the inner retina (Supplementary Figure 1A), PKM2 was found in photoreceptor inner segments and synapses (**Figure 1A**). Expression of mitochondrial cytochrome oxidase (COX) largely overlapped with PKM2 (**Figure 1A**).

Challenging energy metabolism reduces rod and cone photoreceptor viability

To dissect retinal energy metabolism, we selectively manipulated key pathways (**Figure 1B**) using organotypic retinal explants (17) (Supplementary Figure 1B). Retinal cultures were prepared with the RPE cell layer attached to the neuroretina (control) or without RPE (*no* RPE). The *in vivo* expression patterns of enzymes important for energy metabolism (**Figure 1A**) were comparable in the *in vitro* situation (Supplementary Figure 1C). In control, the rate of cell death in the outer nuclear layer (ONL), as assessed by the TUNEL assay (**Figure 1C, D**), was low ($0.46\% \pm 0.22$, $n=9$). The *no* RPE condition displayed significantly increased photoreceptor cell death ($3.35\% \pm 1.62$, $n=5$, $p<0.001$). Blocking RPE glucose transport with the selective GLUT1 inhibitor 1,9-dideoxyforskolin (1,9-DDF) (18) further increased ONL cell death ($6.77\% \pm 1.57$, $n=5$, $p<0.0001$). The inhibition of glycolytic PKM2 with Shikonin (19), and OXPHOS disruption with the electron chain uncoupler carbonyl cyanide-p-trifluoromethoxyphenylhydrazone (FCCP) (16) both caused a significant increase in photoreceptor cell death (Shikonin: $17.75\% \pm 2.13$, $n=5$, $p<0.0001$; FCCP: $22.49\% \pm 6.79$, $n=5$, $p<0.0001$). All four experimental interventions mostly affected photoreceptors, not significantly reducing the viability of other retinal cell types (**Figure 1C**).

Remarkably, a differential effect on cone photoreceptor survival (**Figure 1E, F**) was observed as assessed by immunodetection of the cone-specific marker arrestin-3 (Arr3). Control retina harbored $11.47 (\pm 1.95)$, $n=9$ cones per 100 μm of retinal circumference. *no*RPE retina displayed a significant decrease in cone survival (4.62 ± 1.39 , $n=5$, $p<0.0001$), while blocking RPE glucose transport with 1,9-DDF led to an even more dramatic reduction (1.02 ± 1.09 , $n=5$, $p<0.0001$). By comparison, inhibition of glycolysis with Shikonin had a relatively mild effect on cone survival (7.78 ± 1.36 , $n=5$, $p<0.0001$). Surprisingly, FCCP treatment did not cause cone death (12.84 ± 1.75 , $n=5$, $p=0.22$), compared to control.

We further calculated the percentages of dying cones and rods, assuming that 3% of all photoreceptors were cones (20, 21) (**Figure 1G, H**). In the four treatment situations the ratios of cone to rod death were: *no* RPE = 1:1; 1,9-DDF = 1:1.5; Shikonin = 1:17; and FCCP = 0: ∞ . Cones were almost entirely depleted by the 1,9-DDF treatment but remained virtually unaffected by the FCCP treatment. Rods, however, were strongly and highly selectively affected by Shikonin and FCCP, while the *no* RPE condition had a relatively minor effect. These results highlight important differences between rod and cone energy metabolism.

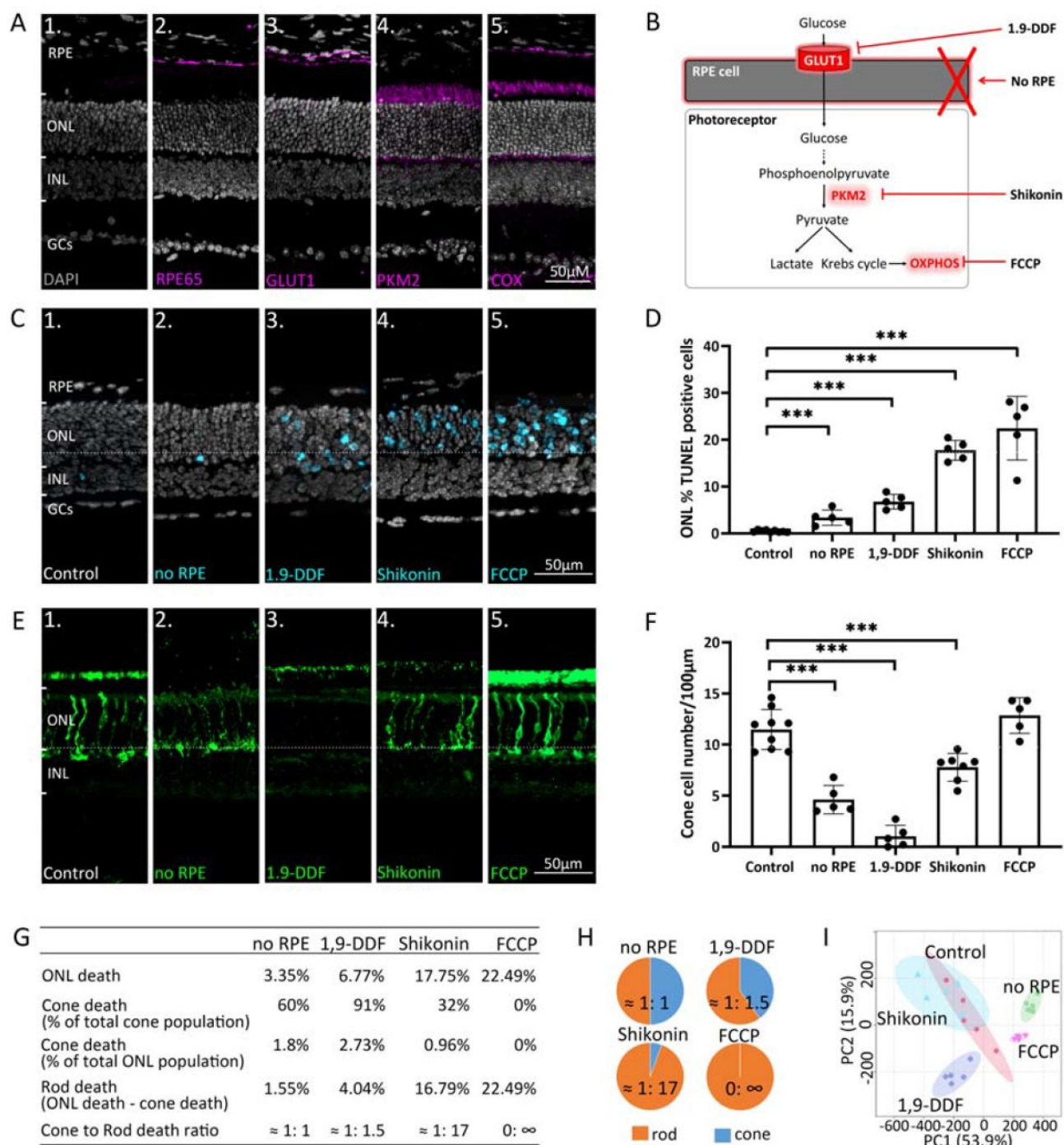


Figure 1.

Manipulating energy metabolism differentially affects rod and cone photoreceptors.

(A) Immunofluorescence staining (magenta): (A1) negative control, (A2) RPE65, (A3) glucose transporter-1 (GLUT1), (A4) pyruvate kinase M2 (PKM2), (A5) mitochondrial cytochrome oxidase (COX). (B) Overview of experimental manipulations. Organotypic retinal explants were cultured with or without RPE or treated with 1,9-DDF, Shikonin, or FCCP. (C) TUNEL assay (cyan) marking dying cells in the five experimental conditions. (D) Quantification of TUNEL-positive cells in outer nuclear layer (ONL). Data represented as mean $\pm$ SD. (E) Cone-arrestin labelling (green) in the ONL. (F) Quantification of arrestin-positive cones. Data represented as mean $\pm$ SD. (G) Table giving percentages of cones and rods lost for each treatment. (H) Pie charts illustrate cone to rod cell death ratios. (I) Principal component analysis (PCA) of retinal samples investigated with ^1H -NMR spectroscopy-based metabolomics. Dots in graphs represent individual retinal explants. Asterisks indicate significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. RPE = retinal pigment epithelium; INL = inner nuclear layer; GCs = ganglion cells.

Experimental retinal interventions produce characteristic metabolomic patterns

We employed high-field (600 MHz) ^1H -NMR spectroscopy-based metabolomics to study the metabolic signatures in five experimental groups (Supplementary Table 1). A principal component analysis (PCA) showed clear group separation (**Figure 1I**). Unbiased clustering of metabolite profiles revealed specific groups that were differentiated among the five experimental situations (**Figure 2A**). The greatest cluster overlap was seen in control and Shikonin groups. Significant clustering was observed in several amino acid sub-classes and energy metabolites. We found the strongest changes in the 1,9-DDF condition, where glutamate, aspartate, alanine, O-phosphocholine, and nicotinamide adenine dinucleotide (NAD^+) concentrations were particularly upregulated compared to control. Further, branched-chain amino acids (BCAAs), lactate, and threonine were upregulated in the FCCP and *no* RPE condition. Glutathione (GSH) and glutathione disulfide (GSSG) were relatively reduced by Shikonin treatment, with increased creatine and formate. Control and 1,9-DDF treatment displayed high levels of guanosine triphosphate (GTP), adenosine triphosphate (ATP), and π -methylhistidine, which were low in the other groups (**Figure 2A**).

We then compared metabolite patterns for each treatment to control (**Figure 2B**). Metabolites were grouped according to class and function, including fundamental metabolic pathways, such as energy and cofactor metabolism, coenzymes, Krebs-cycle, neuronal tissue-associated metabolites and neurotransmitters, BCAAs, glucogenic and ketogenic amino acids (AAs), serine synthesis pathway, cell membrane synthesis, and individual features from redox metabolism. Afterward, the metabolomics differences identified were explored in detail and related to the retinal expression of corresponding enzymes.

Retina cultured without RPE displays strongly increased glycolytic activity

The RPE is tightly bound to the choroid and serves as interface between vasculature and neuroretina. Because of the strong adherence of RPE to the choroid, most prior studies on explanted retina were performed without RPE (4, 11). To understand the metabolic communication between neuroretina and RPE, we prepared organotypic retinal explants both with RPE (control) or without RPE (*no* RPE).

The metabolite profile of control was markedly different from retina cultured without RPE, with 25 metabolites exhibiting significant changes (**Figure 3A**, Supplementary Figure 3). Decreased ATP levels in the *no* RPE group, corresponded to increased lactate. The BCAAs isoleucine, leucine, and valine, and some of the glucogenic/ketogenic AAs threonine, methionine, glycine, displayed higher levels in the *no* RPE group compared to control. Alanine, aspartate, and taurine showed the opposite trend. Hypotaurine, 4-aminobutyrate (GABA), and glutamate were accumulated in retinal tissue, while membrane synthesis-associated metabolites, such as myo-inositol and O-phosphocholine, as well as creatine and N-acetylaspartate (NAA) were reduced in neuroretina. Moreover, higher levels of GSSG appeared in the absence of RPE, pointing towards an altered redox metabolism.

Pattern hunter (Pearson R) correlation analysis found citrate, a central Krebs-cycle metabolite, positively correlated with taurine, alanine, aspartate, and ATP (**Figure 3B**). Negative correlations with citrate included ADP, BCAAs, and lactate. A subsequent KEGG-based pathway analysis revealed changes in arginine and proline metabolism, taurine and hypotaurine metabolism, and alanine, aspartate, and glutamate metabolism in the *no* RPE situation (**Figure 3C**).

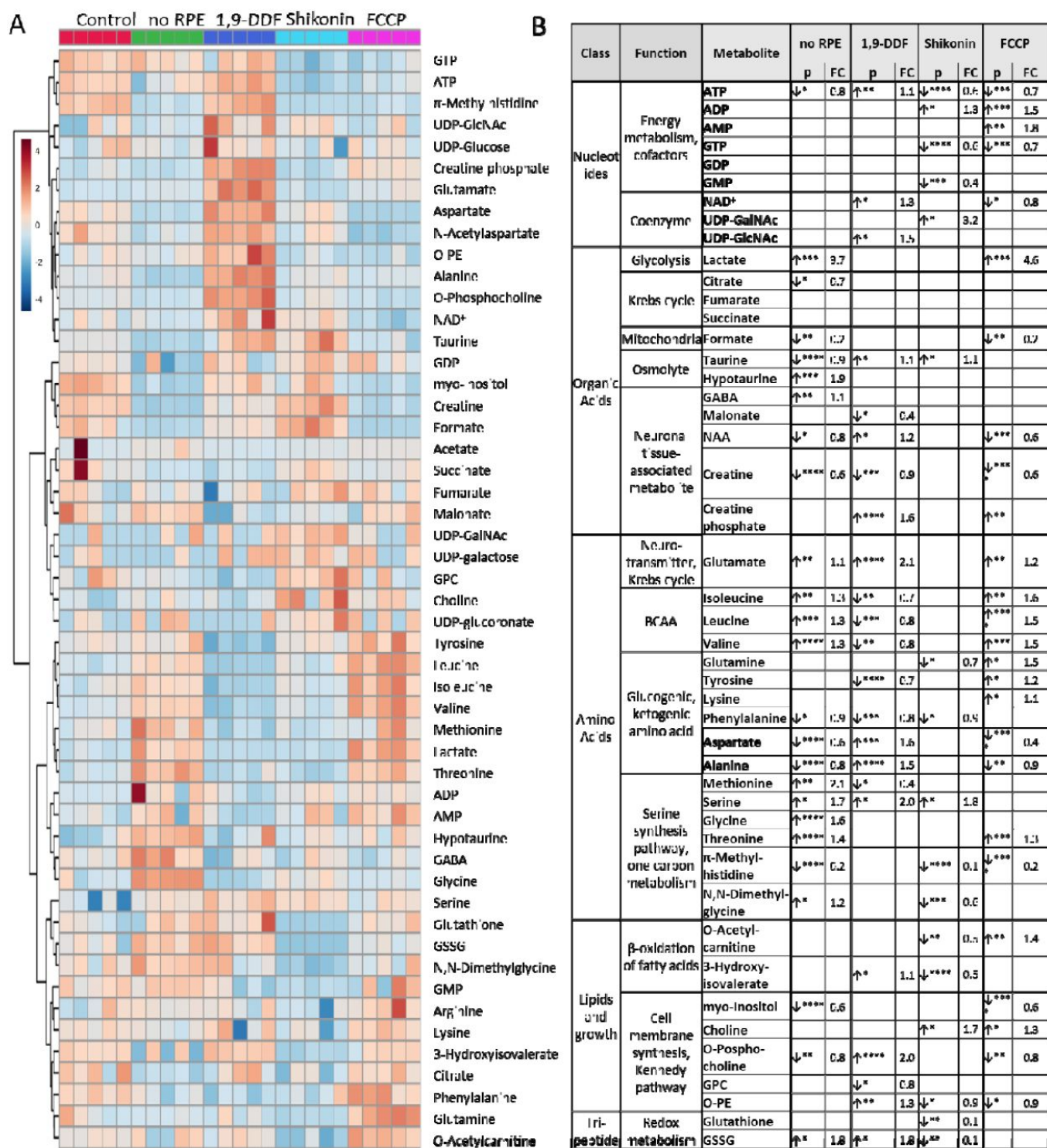


Figure 2.

¹H-NMR spectroscopy-based metabolomic analysis of retina subjected to interventions in energy metabolism.

(A) Heatmap based on unsupervised hierarchical cluster analysis by Ward's linkage, showing auto-scaled metabolite concentrations (red – high, blue – low) in five different experimental conditions: control - red, no RPE - green, 1,9-DDF - dark blue, Shikonin - light blue, FCCP purple (n = 5 samples per condition). (B) Metabolic profiles of each intervention were compared to control. Metabolites significantly changed in at least one experimental condition were grouped according to functions and pathways. Data show p-values and fold change (FC) over control. Statistical comparison: student's unpaired t-test (group variance equal); p-values: ****< 0.0001, ***< 0.001, **< 0.01, *< 0.05. See also Supplementary Figure 2.

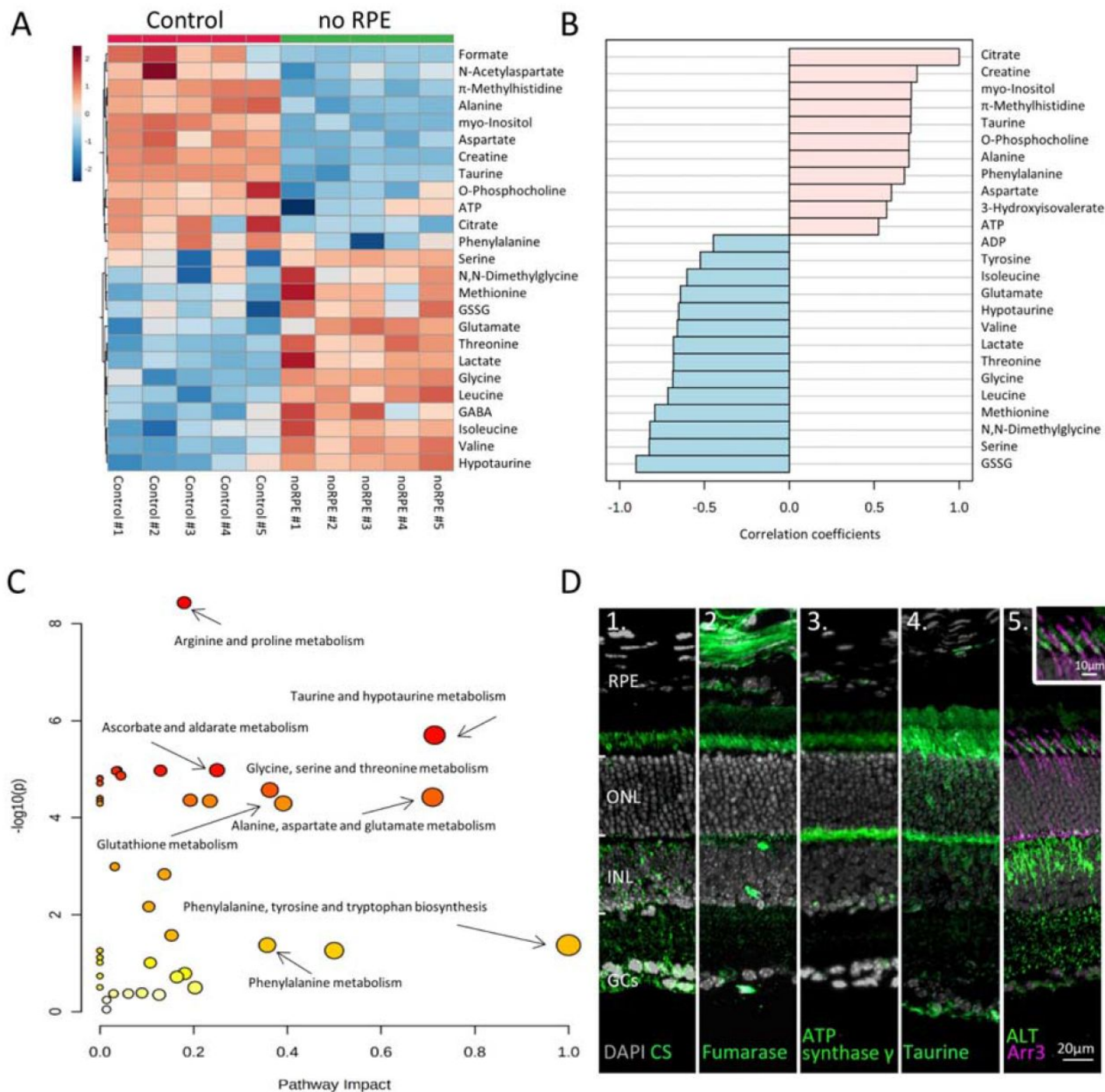


Figure 3.

Absence of RPE dramatically changes retinal metabolism.

(A) Heatmap, based on unsupervised hierarchical cluster analysis by Ward's linkage, illustrating statistically significant changes for 25 metabolites (unpaired t-test, fold change (FC) > 1.2, raw p value < 0.05). (B) Pattern hunter for citrate, showing the top 25 correlating metabolites, correlation coefficient given as Pearson r distance. (C) Metabolic pathways most affected in no RPE condition compared to control, based on KEGG-pathway database. (D) Immunodetection of enzymes and metabolites (green): (D1) citrate synthase (CS), (D2) fumarase, (D3) ATP synthase γ , (D4) taurine, (D5) alanine transaminase (ALT). Co-staining for ALT and cone arrestin (Arr3; magenta) showed high expression in cone inner segments. DAPI (grey) was used as a nuclear counterstain. See also Supplementary Figure 3.

Immunofluorescence was used to localize enzymes likely responsible for the observed metabolic changes (**Figure 3D**). Citrate synthase (CS) and fumarase, key enzymes of the Krebs-cycle, were found mostly in photoreceptor inner segments and synaptic terminals. The γ -subunit of ATP synthase, essential for OXPHOS, was also localized to photoreceptor inner segments and synapses. Thus, low ATP levels in the *no* RPE group likely resulted from decreased photoreceptor Krebs-cycle and OXPHOS activity. High levels of taurine were detected in photoreceptor inner segments and synapses, as well as in the ganglion cell layer (**Figure 3D**). Finally, the enzyme alanine transaminase (ALT), previously thought to be expressed only in muscle and liver, was found to be expressed in the inner nuclear and plexiform layers (**Figure 3D**). ALT was also strongly expressed in cone inner segments, as evidenced by co-labelling with cone -Arr3. This expression pattern and high alanine levels in the control strongly suggested Cahill-cycle activity in the retina (22). When ALT activity was inhibited with β -chloro-alanine, photoreceptor cell death was increased and notably cone survival was reduced, validating the functional expression of ALT in the outer retina (Supplementary Figure 4A, B).

The patterns observed in the *no* RPE to control comparison indicated that retinal metabolism strongly depends on RPE-to-neuroretina interactions, including shuttling of metabolites such as hypotaurine/taurine. Notably, the accumulation of BCAA, glutamate, and lactate, concomitant with decreased alanine levels in the *no* RPE group, implied that RPE removal switched retinal metabolism from Krebs-cycle/OXPHOS to aerobic glycolysis.

Photoreceptors use the Krebs-cycle to produce GTP

We further compared metabolite patterns between *no* RPE and FCCP groups. In **Figure 2**, the *no* RPE and FCCP groups showed similar metabolite patterns and pathway changes overall. However, a detailed statistical analysis revealed 17 significant metabolite changes (**Figure 4A**; Supplementary Figure 5). Among the most highly reduced metabolite concentrations in the FCCP group was GTP, which positively correlated with GABA, ATP, GSSG, hypotaurine, and NAD^+ (**Figure 4B**). In contrast, GDP, BCAAs, glutamine, and citrate negatively correlated to GTP. The KEGG-based pathway analysis ranked glycine, serine and threonine metabolism, glutathione metabolism, and alanine, aspartate and glutamate metabolism as the most significantly changed between the two conditions (**Figure 4C**).

Curiously, measurable amounts of GTP can only be generated in two ways: 1) by succinate-CoA ligase, GTP-forming-1 (SUCLG-1) in the Krebs-cycle, and 2) by nucleoside-diphosphate kinase (NDK) from excess ATP. Since ATP levels were low in both the FCCP and the *no* RPE situation, here, GTP could not have been produced from ATP. Immunofluorescence for SUCLG-1 showed strong expression in photoreceptor inner segments and synapses, where it co-localized to a large extent with mitochondrial COX (**Figure 4D**). Hence, the SUCLG-1 retinal expression pattern and the synthesis of GTP in the absence of RPE provided clear evidence for Krebs-cycle activity in photoreceptors.

We also identified an FCCP-induced upregulation of citrate, concomitant with a downregulation of NAD^+ . This indicated a Krebs-cycle interruption at the level of D-isocitrate to α -ketoglutarate conversion, *i.e.*, a step that requires NAD^+ . NAD^+ is known to be consumed by poly-ADP-ribose-polymerase (PARP), which is activated by oxidative DNA damage (23) and up-regulated in dying photoreceptors (24). Therefore, we quantified poly-ADP-ribose (PAR) accumulation in retinal cells as a marker for PARP activity. Compared to the *no* RPE group, more photoreceptor cells in the FCCP-treated retina showed PAR accumulation, correlating with decreased retinal NAD^+ levels (**Figure 4E, F**). Hence, FCCP-induced oxidative stress may cause increased PARP activity and decreased NAD^+ levels, eventually interrupting the Krebs-cycle, as evidenced by citrate accumulation.

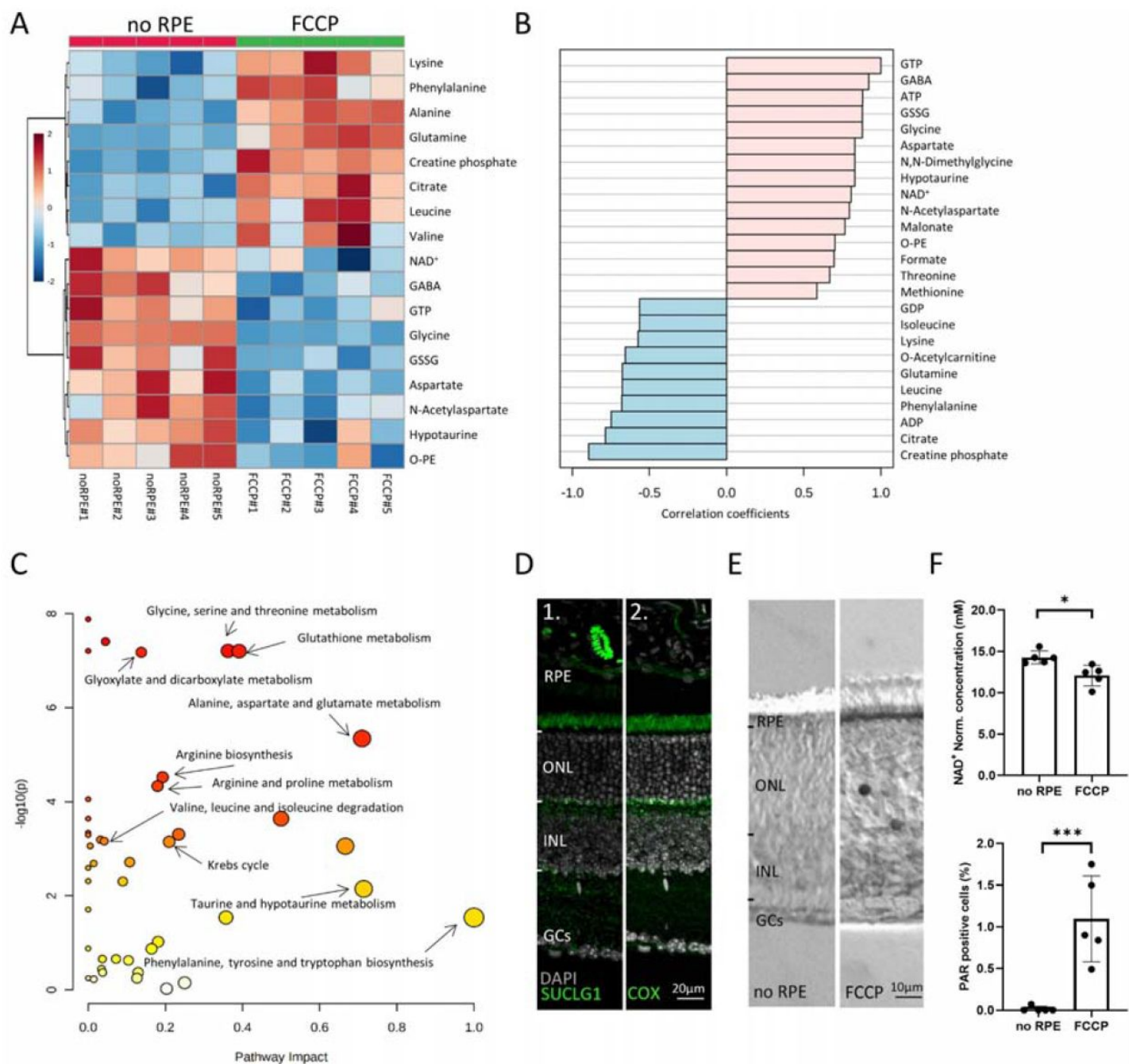


Figure 4.

Comparison between *no RPE* and *FCCP* conditions.

(A) Heatmap, based on unsupervised hierarchical cluster analysis by Ward's linkage, illustrating statistically significant metabolite changes (unpaired t-test, fold change > 1.2, raw p-value < 0.05). (B) Pattern hunter for guanosine triphosphate (GTP) showing the top 25 correlating compounds. Correlation coefficient given as Pearson *r* distance. (C) KEGG-based pathway analysis, comparison between *no RPE* and *FCCP*. (D) Immunofluorescence for succinate-CoA ligase-1 (SUCLG1, green) labelled photoreceptor inner segments and colocalized with COX. DAPI (grey) was used as a nuclear counterstain. (E) PAR positive photoreceptors (black) in the outer nuclear layer (ONL) of *FCCP* treated retina. (F) Compared to *no RPE*, NAD⁺ levels were lower in the *FCCP* group, while the percentage of PAR positive cells was higher. Data represented as mean $\pm$ SD. p values: **** < 0.001, * < 0.05. See also Supplementary Figure 5.

Overall, the data from the *no RPE* to FCCP group comparison showed that disruption of OXPHOS led to AA accumulation, including lysine, phenylalanine, glutamine, leucine, and valine. Notably, alanine accumulation in the FCCP group was likely caused by ALT-dependent pyruvate transamination. By contrast, metabolites high in the *no RPE* group but low with FCCP treatment were probably generated via the Krebs-cycle in the neuroretina. This concerned especially GTP, aspartate, and NAA.

Reduced retinal glucose uptake promotes anaplerotic metabolism

Since GLUT1 was predominantly expressed in the RPE, we initially assumed that the metabolic response to 1,9-DDF treatment might resemble that of the *no RPE* situation. However, the patterns found between these two groups were very different (*cf.* [Figure 2](#)) revealing 28 significantly changed metabolites ([Figure 5A](#); Supplementary Figure 6). When compared to the *no RPE*, the 1,9-DDF group showed changes in mitochondria and Krebs-cycle-associated metabolites (*e.g.*, high ATP, high formate, low BCAA). Conversely, the Cahill-cycle product alanine and NAA were upregulated by 1,9-DDF treatment. The pattern hunter analysis showed that metabolites formate, glutamate, taurine, and citrate were positively correlated with ATP, while, AMP, GMP, and BCAA were negatively correlated with ATP ([Figure 5B](#)).

The KEGG pathway analysis showed alanine, aspartate, and glutamate metabolism, glutathione metabolism and glycine, serine, and threonine metabolism among the three most affected pathways (Supplementary Figure 6B). Taken together, the depletion of BCAAs in the 1,9-DDF treatment indicated that the retina used anaplerotic substrates to fuel the Krebs-cycle and maintain ATP production.

Inhibition of glycolysis strongly impacts retinal ATP production

We compared the Shikonin treatment, which inhibits the last step of glycolysis, to the FCCP treatment, which abolishes mitochondrial ATP synthesis ([Figure 5C](#)). Formate, aspartate, and NAA were increased in the Shikonin group. Conversely, higher amounts of GSH, GSSG, BCAA, threonine, and phenylalanine after FCCP treatment indicated Krebs-cycle interruption (Supplementary Figure 7). The production of lactate was higher with FCCP-compared to Shikonin treatment, reflecting increased glycolysis to compensate for the loss of Krebs-cycle-dependent ATP production. Metabolites connected to oxidative stress, such as GSSG and GSH, were also increased by FCCP treatment. Pattern hunter for metabolites correlated with glutathione found 3-hydroxyisovalerate, glutamine, GSSG, but also lactate and ATP, while succinate, alanine, and GABA were negatively correlated with GSH ([Figure 5D](#)). The subsequent pathway analysis identified arginine biosynthesis, alanine, aspartate, and glutamate metabolism, as well as glycolysis and gluconeogenesis as the most strongly regulated metabolic pathways (Supplementary Figure 7B).

Blocking glucose uptake and OXPHOS uncoupling reveal distinct rod and cone metabolism

The metabolic rate of cones is at least two times higher than that of rods ([9](#)). The treatment with 1,9-DDF resulted in near complete cone loss, while cones were fully preserved with FCCP treatment (*cf.* [Figure 1](#)). Therefore, we compared these two experimental groups with control to investigate the relative contributions of rods and cones to the metabolite patterns observed.

In this three-way comparison, 29 metabolite concentrations were significantly changed ([Figure 6A](#); Supplementary Figure 8). The FCCP treatment reduced ATP and GTP levels, while lactate production was greatly increased, along with methionine and threonine. BCAAs, glutamine, phenylalanine and tyrosine also showed a significant upregulation with FCCP treatment. Interestingly, some metabolites like GSSG, serine, glutamate, alanine, NAD^+ , and taurine displayed a pronounced increase in the 1,9-DDF group when compared to either control or FCCP. In addition, formate, aspartate, and myo-inositol were upregulated with 1,9-DDF treatment, together with NAA

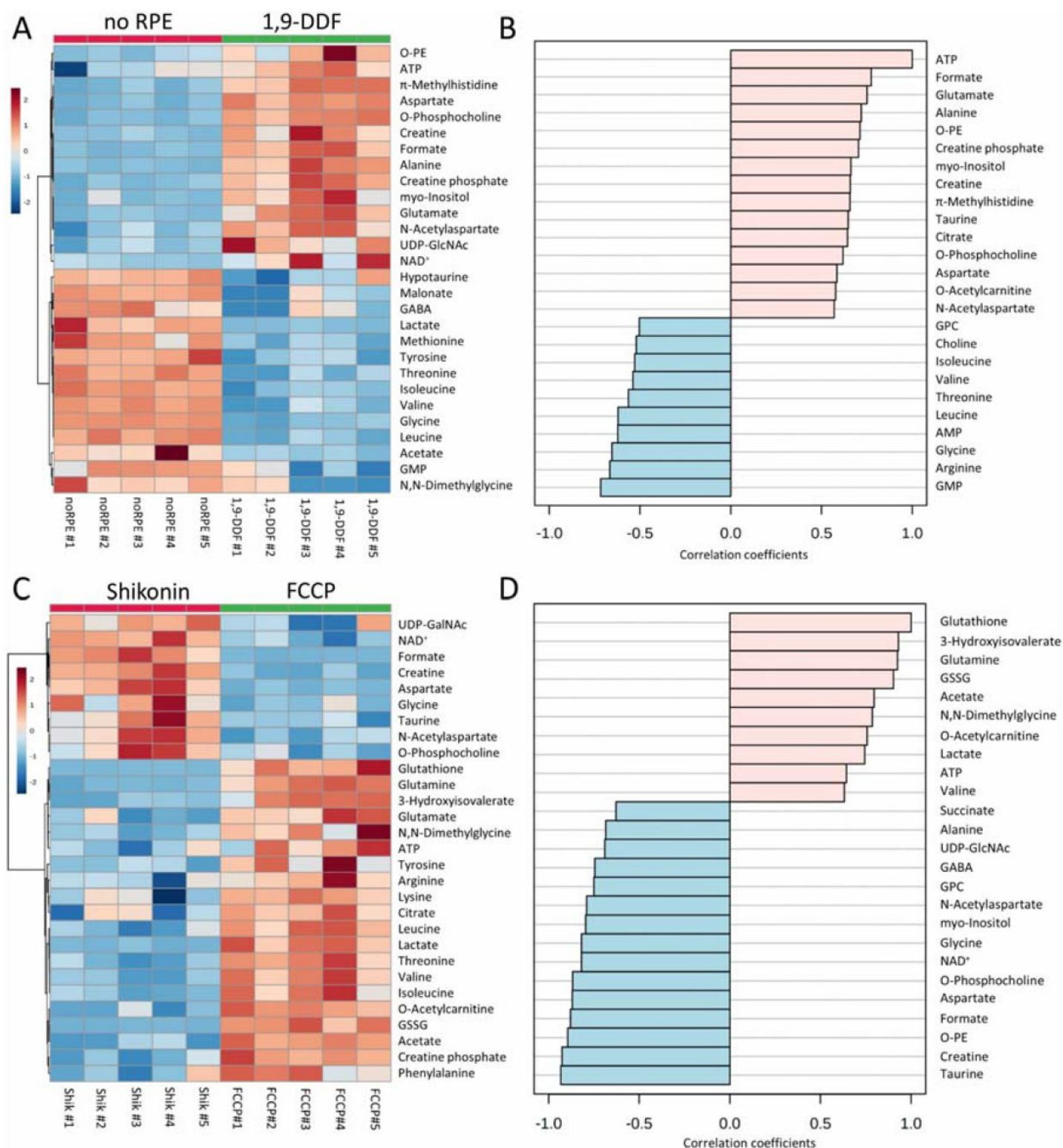


Figure 5.

Metabolomic analysis of no RPE vs. 1,9-DDF treatment and Shikonin vs. FCCP treatment.

Heatmap illustrating statistically significant metabolite changes (unpaired t-test, fold change (FC) > 1.2, raw p value < 0.05) for (A) no RPE vs. 1,9-DDF and (C) Shikonin vs. FCCP. Pattern hunter for (B) ATP and (D) glutathione, showing the top 25 correlating compounds. Correlation coefficient given as Pearson r distance. See also Supplementary Figures 6 and 7.

and creatine. Finally, the comparison between the 1,9-DDF and FCCP groups revealed highly increased levels of ATP and GTP, implying a strong activation of the Krebs-cycle under 1,9-DDF treatment.

A cell type-specific attribution of these metabolite patterns may be superseded by direct drug treatment effects. Using only metabolite changes in opposing directions from control to interpret different cellular compositions, we identified BCAAs, tyrosine, NAA, and ATP (Supplementary Figure 8). BCAAs and tyrosine (and glutamine) exhibited low levels in the rod-rich 1,9-DDF group and high levels in the cone-rich FCCP group, indicating that these metabolites might be consumed by rods or produced in cones. Vice versa, it can be assumed that aspartate and NAA were predominantly produced in rods.

To investigate the role of glutamine in retinal metabolism, we used immunofluorescence (**Figure 6B**). We confirmed a strong expression of glutamine synthase (GS) in MGCs (25). Remarkably, the enzyme that hydrolyses glutamine to glutamate, glutaminase C (GAC) was prominently expressed in photoreceptor inner segments, photoreceptor synapses, and INL. Co-labelling with cone Arr3 revealed a strong expression of GAC in cone inner segments, implying that cones use glutamine imported from MGCs as a metabolic substrate. Accordingly, treatment with the GAC-specific inhibitor Compound 968 dose-dependently reduced photoreceptor viability, while leaving the inner retina mostly intact (Supplementary Figure 4C, D).

Impact on glycolytic activity, serine synthesis pathway, and anaplerotic metabolism

To assess how the various experimental interventions affected retinal metabolism, we calculated the ratios for pathway-specific metabolites (**Figure 6C**). GTP, when produced by SUCLG-1, is a marker for Krebs-cycle activity, while lactate indicates glycolytic activity. In the *no* RPE group, the GTP-to-lactate ratio was significantly lower than in control, indicating 4.1-times higher glycolytic activity. While in the 1,9-DDF and Shikonin group the GTP-to-lactate ratios were not significantly different from control, with FCCP treatment this ratio dropped to the lowest level, in line with a strong downregulation of Krebs-cycle and concomitant upregulation of glycolysis.

We then calculated the ratio of serine to lactate as an indicator for serine synthesis pathway activity. Under control conditions, and in the *no* RPE and FCCP groups, this activity was rather low, while it was strongly increased by 1,9-DDF and Shikonin treatment. In the Shikonin group, serine production from 3-phosphoglycerate and subsequent deamination to pyruvate may bypass the PKM2 block. Serine is also a precursor of phosphatidylserine, one of the three main cell membrane components. Together with a reduction of choline and high concentrations of myo-inositol, O-phosphocholine, and O-phosphoethanolamine, this may reflect high cell membrane synthesis under 1,9-DDF treatment.

Finally, we used the ratio of GTP to BCAAs to investigate to what extent Krebs-cycle activity was driven by anaplerotic metabolism. A comparatively low GTP to BCAA ratio in the *no* RPE situation showed that much of the GTP generated in the neuroretina likely came from anaplerotic substrates. While both Shikonin and FCCP treatment decreased BCAA use, 1,9-DDF treatment increased anaplerotic metabolism.

Evidence for mini-Krebs-cycle activity

Comparing the ratios of GTP to lactate with GTP to BCAA in the *no* RPE situation showed that a large proportion of the GTP produced in the neuroretina by SUCLG-1 was derived from anaplerotic substrates rather than from pyruvate. These substrates enter the Krebs-cycle mostly at the level of α -ketoglutarate, suggesting that the neuro-retinal Krebs-cycle might not start with citrate. Instead, only an α -ketoglutarate to oxaloacetate shunt would be employed, including the GTP-synthesizing step from succinyl-CoA to succinate (**Figure 7A**). A key step of this mini-Krebs-

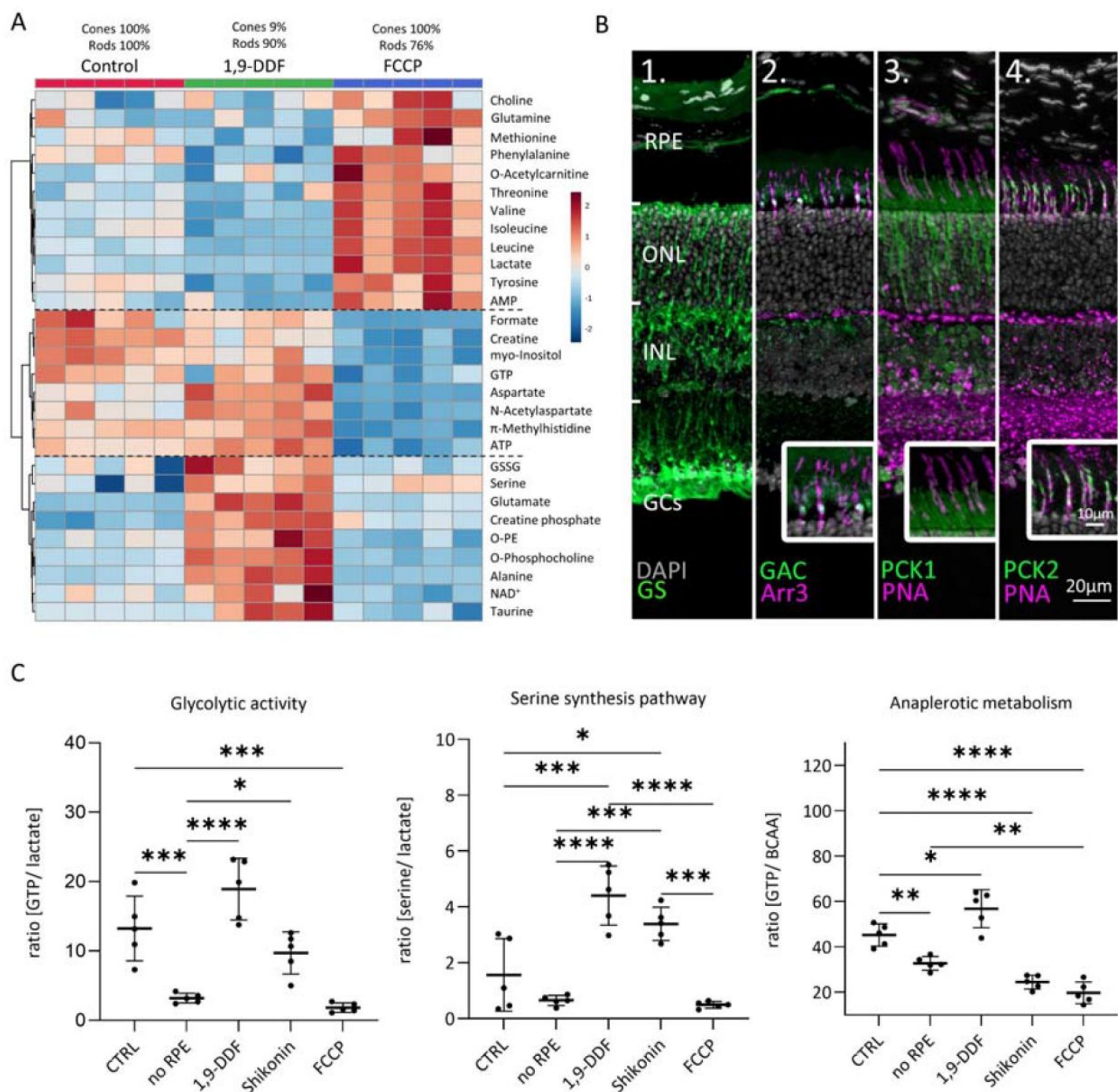


Figure 6.

Metabolomic comparison between control, 1,9-DDF, and FCCP treatment.

(A) Heatmap illustrating 29 statistically significant metabolite changes (parametric one-way ANOVA, Fisher's LSD post-hoc analysis). Three main clusters of metabolite changes were evident (dashed lines). (B) Immunostaining (green) for enzymes related to glutamine metabolism. DAPI (grey) was used as nuclear counterstain. (B1) Glutamine synthase (GS), (B2) glutaminase C (GAC); co-localization with cone-arrestin (Arr3; magenta). (B3) phosphoenolpyruvate carboxykinase 1 (PCK1) and (B4) PCK2, both co-localized with cone marker peanut-agglutinin (PNA). (C) Ratios between metabolites representing glycolysis (lactate vs. GTP), anaplerotic metabolism (GTP vs. BCAA), and serine synthesis pathway (serine vs. lactate). Data represented as individual data points with mean $\pm$ SD. Statistical comparison using one-way ANOVA, Tukey's multiple comparisons test; p values: **** $<$ 0.0001, *** $<$ 0.001, ** $<$ 0.01 * $<$ 0.05. See also Supplementary Figure 8.

cycle (28 [↗](#)) is the transamination of oxaloacetate with glutamate by aspartate aminotransferase (AAT) to yield α -ketoglutarate and aspartate, which is further acetylated to NAA. To investigate this possibility, we analyzed eight metabolites associated with the Krebs-cycle (citrate, succinate, fumarate, GTP) and hypothesized mini-Krebs-cycles (glutamine, glutamate, aspartate, NAA).

Hierarchical clustering showed similar patterns for aspartate and NAA in all experimental conditions (**Figure 7B** [↗](#)). Except for the FCCP treatment, the ratio of citrate/NAA was 0.5, indicating that the retina preferred the mini-Krebs-cycle over the full Krebs-cycle (**Figure 7C** [↗](#)). Immunolabelling for AAT – conventionally associated with muscle and liver metabolism (29 [↗](#), 30 [↗](#)) – found both AAT1 and AAT2 to be expressed in photoreceptor inner segments and cell bodies (**Figure 7D** [↗](#)). This confirmed that photoreceptors can execute the mini-Krebs-cycle, while the NAA production seen in the metabolomic data demonstrated that this cycle was indeed used.

Finally, we compared the energetic efficiencies of glycolysis, Krebs, Cori, Cahill, and mini-Krebs cycle (**Figure 8** [↗](#)). Using anaplerotic substrates the Cahill and mini-Krebs cycle allows for far more efficient energy production compared to the Cori-cycle. Moreover, mitochondrial PCK2 can regenerate pyruvate from oxaloacetate. Assuming that one mole of NADH/FADH₂ can be used to generate three moles of ATP via OXPHOS, and depending on the exact stoichiometry of glutamate/BCAA input, the mini-Krebs-cycle may deliver up to 18 moles of ATP and 2 moles of GTP per 3 carbon unit. Astonishingly, this exceeds the 15 moles of ATP and 1 mole of GTP generated in the “full” Krebs-cycle.

Discussion

This study combined cellular enzyme expression patterns with quantitative metabolomics of retinal explants. In an entirely controlled environment, this enabled a comprehensive assessment of retinal metabolism and allowed to analyze the impact of different experimental conditions upon cell death and survival of retinal cells. As opposed to earlier works, our study highlights the crucial importance of OXPHOS and anaplerotic pathways for the maintenance and survival of rod and cone photoreceptors. Importantly, we show that photoreceptors can uncouple glycolysis and Krebs-cycle using a NAA-producing shunt, allowing optimal function of both pathways and solving a long-standing problem in neuronal energy metabolism research. Because of the ramifications for overall cellular physiology, these findings are highly relevant for future therapy developments, for retinal diseases, as well as for neurodegenerative and further metabolic diseases in general.

Photoreceptors use glucose and anaplerotic substrates as fuel

Under anaerobic conditions, the production of lactate from glucose is a hallmark of skeletal muscle function. Here, pyruvate is reduced to lactate, which enters the bloodstream to serve as substrate for liver gluconeogenesis. Glucose can then cycle back to the muscle. This glucose-lactate cycle between muscle and liver was discovered in the 1930s and is referred to as Cori-cycle (32 [↗](#)). The release of large amounts of lactate from isolated retina suggested that also the retina might use the Cori-cycle, albeit under aerobic conditions (33 [↗](#)). Photoreceptors have thus been proposed to consume primarily glucose via aerobic glycolysis (11 [↗](#)). The resultant lactate would be used as fuel in the Krebs-cycle by RPE and MGCs, generating ATP via OXPHOS. However, this hypothesis is contradicted by the high density of mitochondria in photoreceptor inner segments (26 [↗](#)). Moreover, key enzymes for Krebs-cycle (CS, FH, SUCLG1) and OXPHOS (COX, ATP synthase γ) were strongly expressed in photoreceptors, but showed little or no expression in RPE or MGCs, in line with previous research (34 [↗](#)). When we cultured retina without RPE, as done by Warburg and others (4 [↗](#), 11 [↗](#)), we confirmed a strong lactate production. Yet, in retina cultured with RPE, lactate production was minor. Moreover, retina without RPE still produced large amounts of GTP, likely stemming from SUCLG1 activity and demonstrating Krebs-cycle operation in photoreceptors. Neuroretina without RPE displayed reduced viability of both rod and cone photoreceptors, as well

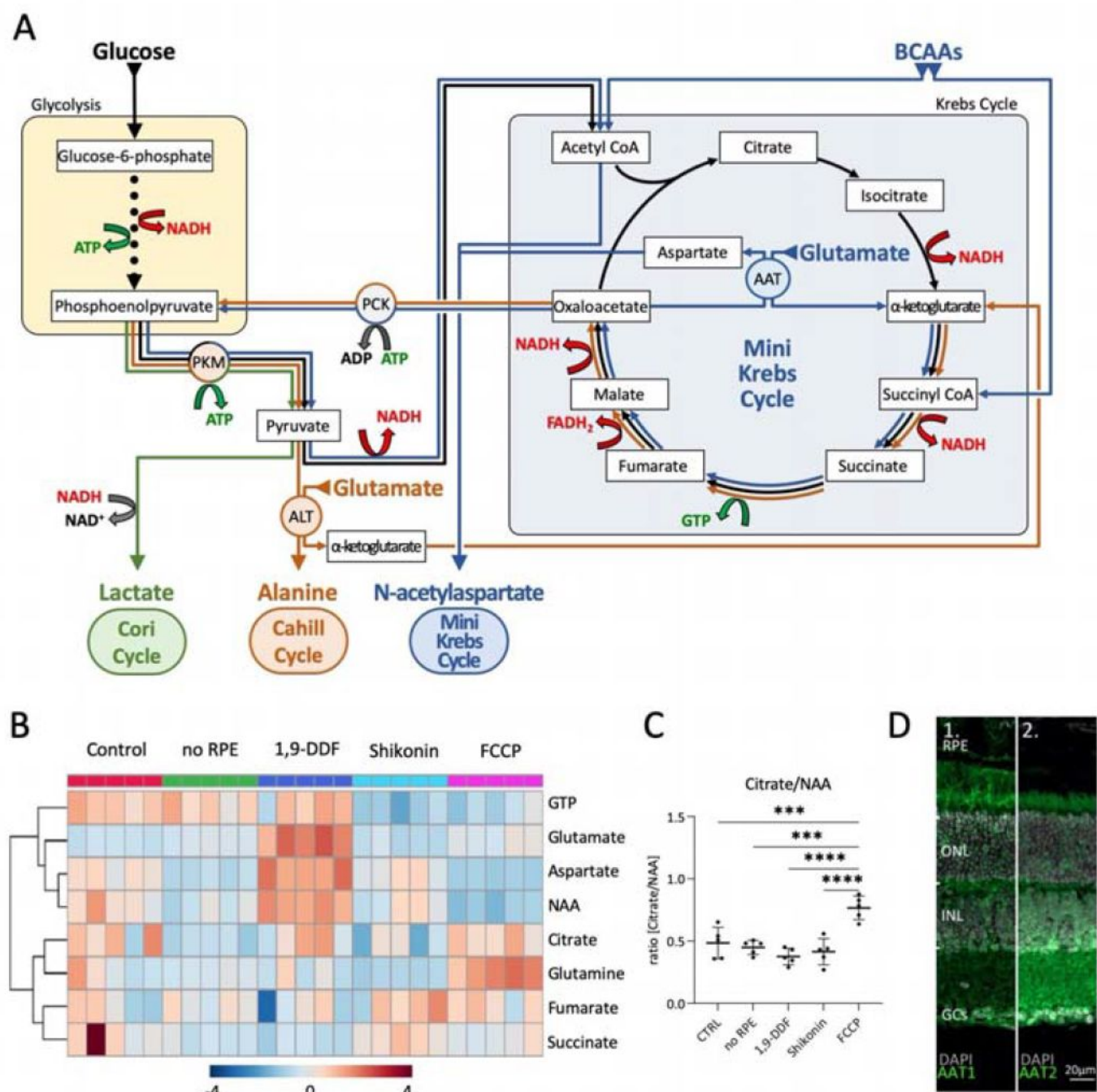


Figure 7.

Metabolic pathways in the retina, key metabolites, and expression of aspartate amino transferase (AAT).

(A) Overview of main metabolic pathways and metabolites. Execution of Cori- (green arrows), Cahill- (brown), or mini-Krebs-cycle (blue) releases the signature metabolites lactate, alanine, and N-acetylaspartate (NAA), respectively. Key enzymes of the Cahill- and mini-Krebs-cycle are phosphoenolpyruvate carboxykinase (PCK), pyruvate kinase M (PKM), alanine transaminase (ALT) and AAT. (B) Hierarchical clustering of eight metabolites connected to Krebs and mini-Krebs cycle. (C) Ratio NAA vs. citrate, representing full and mini-Krebs-cycle. Data represented as ratio of individual data points with mean $\pm$ SD. (D) AAT-1 and -2 staining (green) with DAPI (grey) as nuclear counterstain.

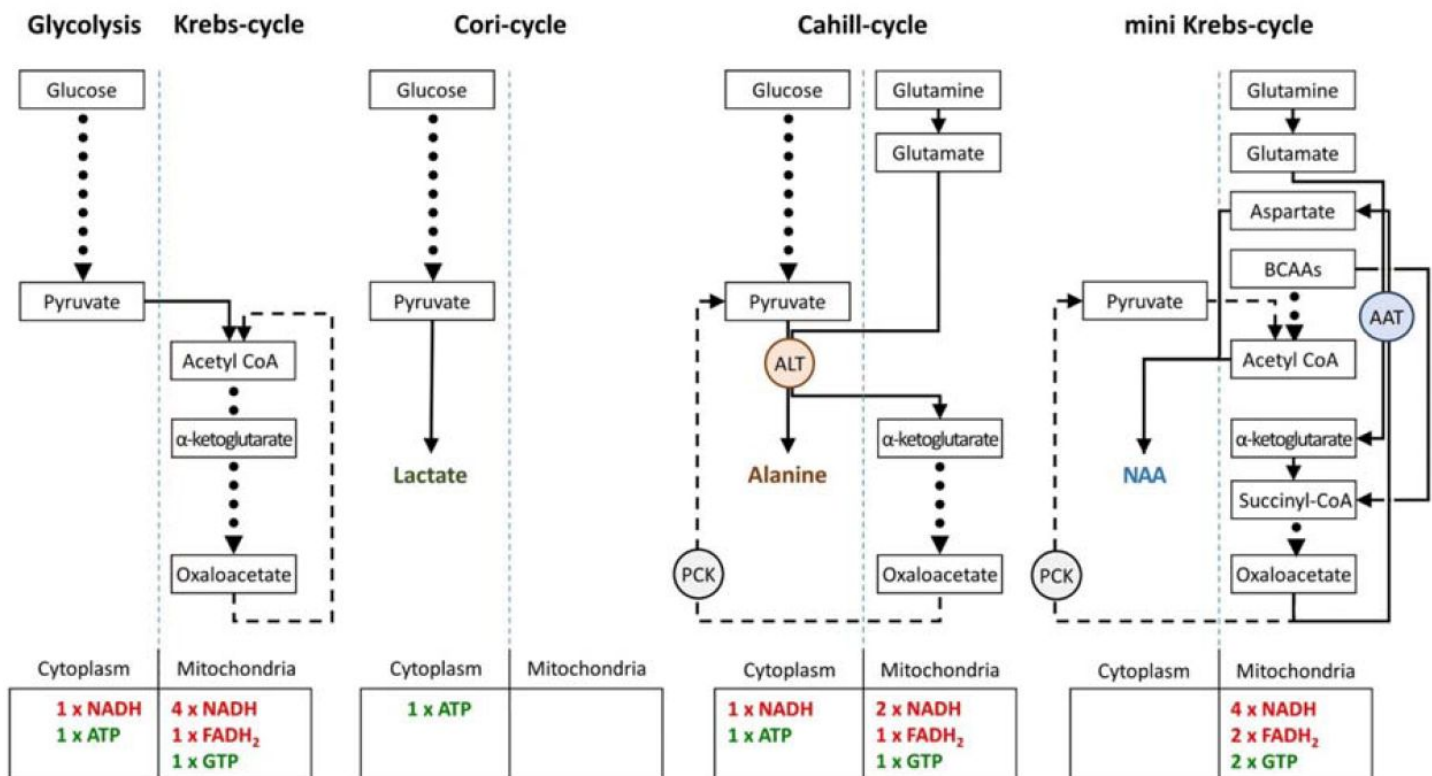


Figure 8.

Comparison of metabolic pathways and their energetic efficiencies.

Compared to the Cori-cycle, both Cahill- and mini-Krebs-cycles are highly efficient. Their key enzymes – alanine transaminase (ALT) and aspartate amino transferase (AAT), respectively – generate alanine and aspartate/N-acetylaspargate (NAA). Pyruvate carboxy kinase (PCK), either in cytoplasm or within mitochondria, may reconstitute pyruvate from oxaloacetate. Note that energy output of each pathway was calculated based on input of pyruvate (three carbons) or glutamate/branched-chain-amin-acid/acetyl-CoA (three carbons).

as an accumulation of BCAAs and lactate. In contrast, the block of glucose import into the RPE by treatment with 1,9-DDF (18 [↗](#)) resulted in a depletion of BCAAs and other AAs, and an increase of retinal ATP, demonstrating a switch from glycolytic to anaplerotic metabolism (28 [↗](#)). Hence, we show that photoreceptors consume both glucose and anaplerotic substrates as fuels, and that retinal metabolism switches to aerobic glycolysis only in the absence of RPE. This switch is likely driven by increased glucose uptake. Normally, tight junction-coupled RPE cells form the outer blood-retinal barrier and prevent direct access of glucose to photoreceptors (35 [↗](#)). In the absence of RPE, photoreceptors expressing the high affinity/high capacity GLUT3 become “flooded” with glucose. The resultant Crabtree effect then causes a shutdown of Krebs-cycle activity (14 [↗](#), 36 [↗](#), 37 [↗](#)). Taken together, the high rates of retinal aerobic glycolysis first reported by Otto Warburg are likely an artefact of the absence of RPE, while intact retina may rely more on OXPHOS for its energy production.

Rods need OXPHOS, cones need glycolysis

Treatment with FCCP revealed striking differences between rod and cone energy metabolism. FCCP eliminates the proton gradient between inner mitochondrial membrane and mitochondrial matrix, abolishing ATP synthesis (16 [↗](#)). Initially, FCCP may increase Krebs-cycle activity to attempt restoration of mitochondrial proton gradient (38 [↗](#)), increasing oxidative stress (39 [↗](#)) and oxidative DNA damage. The resultant activation of PARP would deplete NAD^+ , further aggravating metabolic stress (40 [↗](#)). Although FCCP had a strong toxic effect on rod photoreceptors, remarkably, cone photoreceptors were almost completely preserved. In striking contrast to the FCCP effect was that of the 1,9-DDF block on GLUT1 (18 [↗](#)). Although GLUT1 was strongly expressed in the RPE, glucose entering the RPE is shuttled forward to photoreceptors (10 [↗](#)). Remarkably, with 1,9-DDF, over 90 % of cones were lost, while the detrimental effect on rods was comparatively minor. These differential effects of FCCP and 1,9-DDF strongly suggest that glycolysis is sufficient and necessary for cone survival, in agreement with earlier studies (41 [↗](#), 42 [↗](#)). Conversely, rods require OXPHOS for their survival, while glycolysis is of minor importance. Furthermore, cone viability relative to rods was also strongly compromised in the *no* RPE situation, indicating that an intact blood-retinal barrier (35 [↗](#)) and regulation of glucose access was important for cone survival. We note that GLUT2 may also contribute to photoreceptor glucose uptake (43 [↗](#)).

Cone photoreceptors use the Cahill-cycle

The lactate-generating Cori-cycle (32 [↗](#)) is highly inefficient and in intact retina likely plays a lesser role. An efficient pathway is the glucose-alanine cycle, or Cahill-cycle, in which pyruvate, instead of being reduced to lactate, is transaminated to alanine (22 [↗](#)). This preserves NADH and generates α -keto acids to fuel the Krebs-cycle. The alanine generated enters the bloodstream and is taken up by the liver, where the ammonia is excreted in the form of urea, while the carbon backbone is used for gluconeogenesis. The key enzyme for the Cahill-cycle is ALT, conventionally associated with muscle and liver (29 [↗](#), 30 [↗](#)). Previously, ALT was found in glial cells of the honeybee retina (44 [↗](#)), and ALT activity was detected in rat retinal tissue lysates (45 [↗](#)). In our study, the localization of ALT in photoreceptors and inner retinal neurons, combined with our interventional and metabolomic datasets, demonstrated the operation of the Cahill-cycle in the mammalian retina. Moreover, the expression of mitochondrial PCK2 in cones facilitates the efficient uncoupling of glycolysis from the Krebs-cycle, suggesting that cones may use the Cahill-cycle for very effective energy production.

Glutamine from MGCs provides additional fuel for cones

MGCs are known for their uptake of extracellular glutamate and use for glutamine synthesis. In fact, in retinal histology, glutamate-aspartate transporter (GLAST) and glutamine synthase (GS) have been widely used as markers for MGCs (25 [↗](#), 34 [↗](#)). Glutaminase C (GAC) converts glutamine back to glutamate, which may then serve as a substrate for the mini-Krebs-cycle. We

localized GAC in inner retinal neurons and photoreceptors, with a particularly strong expression in cone inner segments. Here, GAC overlaps with ALT expression, indicating that cones can use glutaminolysis to obtain extra glutamate for pyruvate transamination in the Cahill-cycle. This may also explain why the glycolysis inhibitor Shikonin reduced cone viability much less than that of rods.

A hypotaurine-aurine shuttle can transfer NADH from RPE to photoreceptors

The regeneration of the photoreceptor photopigment retinal is performed by the RPE (12 [↗](#)). Recently, retinal has been proposed to form a Schiff base adduct with taurine, which would act as a retinal carrier and buffer (46 [↗](#)). We found that retina cultured with RPE harbored high levels of taurine and low levels of hypotaurine, while retina without RPE displayed low levels of taurine and high levels of hypotaurine. Our taurine immunostaining found essentially no taurine in the RPE, while photoreceptor inner segments and synapses displayed very high taurine levels. Together, these findings suggest a hypotaurine-aurine shuttle between RPE and photoreceptors. In the RPE taurine can be reduced to hypotaurine and shuttled back to photoreceptors where oxidation by hypotaurine dehydrogenase (46 [↗](#), 47 [↗](#)) may reconstitute taurine, yielding additional NADH for OXPHOS. The net effect of this hypotaurine-aurine shuttle would be a transfer of reducing power from RPE to photoreceptors, boosting photoreceptor ATP production via OXPHOS.

Rods use the mini-Krebs-cycle to uncouple glycolysis from mitochondrial metabolism

A key problem in the understanding of cellular energy metabolism is the relationship between fast but inefficient glycolysis and slow but efficient Krebs-cycle/OXPHOS (48 [↗](#), 49 [↗](#)). Both pathways are coupled via pyruvate and the different metabolic flow rates reduce the efficiency of energy production, *e.g.*, via feedback inhibition (50 [↗](#)). Pyruvate coupling is especially problematic in high and rapidly changing energy demand such as in neurons and photoreceptors (51 [↗](#)). Uncoupling glycolysis and Krebs-cycle via the Cori-cycle is extremely wasteful and likely insufficient to satisfy long-term photoreceptor energy demand. By comparison, the Cahill-cycle delivers additional NADH when using pyruvate derived from glycolysis, and cones may use the Cahill-cycle for uncoupling from glycolysis. An alternative pathway is the mini-Krebs-cycle, essentially an oxaloacetate to α -ketoglutarate shunt (28 [↗](#)). This cycle uses glutamate, glutamine, and BCAAs as fuels to run mitochondrial respiration independent of glycolysis. The key step of the mini-Krebs-cycle is the transamination of oxaloacetate/glutamate to aspartate/ α -ketoglutarate by AAT. α -Ketoglutarate is then metabolized to oxaloacetate, generating NADH, FADH_2 , and GTP. Acetyl-CoA generated from BCAAs or pyruvate is used to create NAA, the end product of the mini-Krebs-cycle. We found mitochondrial AAT2 to be expressed in rod inner segments, in agreement with an early cytochemical study (52 [↗](#)). This indicates that the mini-Krebs-cycle is used primarily by rods, explaining their selective vulnerability to FCCP treatment. Crucially, the mini-Krebs-cycle is more energy-efficient than the Cahill-cycle and generates NAA instead of alanine as net product. Both metabolites serve the purpose of disposing of excess ammonia originating from AA input (53 [↗](#), 54 [↗](#)). NAA in the human brain is one of the most abundant metabolites and is routinely used in clinical MRI diagnosis to visualize brain health (55 [↗](#)). While different functions have been hypothesized for NAA (56 [↗](#), 57 [↗](#)), our work proposes NAA as a signature metabolite for the mini-Krebs-cycle. Indeed, a recent study using Raman spectroscopy imaging detected high levels of NAA in the human retina, demonstrating *in vivo* use of this cycle (58 [↗](#)). Importantly, the 6-step mini-Krebs-cycle is significantly faster than the 10-step Krebs-cycle (28 [↗](#)), more energy-efficient, and uncoupled from glycolysis. In rods, PCK1 may allow to replenish cytoplasmic acetyl-CoA pools from oxaloacetate, if BCAA-derived input was insufficient.

Overall, the different pathways outlined here provide photoreceptor cells with remarkable versatility and flexibility, notably in terms of substrate use, allowing them to dynamically adapt the timing and quantitative output of energy metabolism. Rods rely strongly on the Krebs-cycle and OXPHOS, while cones are more dependent on glycolysis. These differences between rod and cone metabolism may be related to their response kinetics and sensitivities. The flexible uncoupling of glycolysis from mitochondrial respiration, allows both processes to run at optimum, producing the characteristic signature metabolites lactate (Cori-cycle), alanine (Cahill-cycle), and NAA (mini-Krebs-cycle). These metabolites can reveal energy or disease status and could serve as readout and guide for the design of novel therapeutic interventions. Given the general importance of energy metabolism, the significance of our findings extends beyond the retina, for instance, to other neurodegenerative and metabolic diseases.

Materials and Methods

C3H wild-type (WT) mice were used to generate organotypic retinal explant cultures (18¹). All efforts were made to minimize the number of animals used and their suffering. Protocols compliant with the German law on animal protection were reviewed and approved by the “Einrichtung für Tierschutz, Tierärztlichen Dienst und Labortierkunde” of the University of Tübingen and were following the association for research in vision and ophthalmology (ARVO) statement for the use of animals in vision research. For details on the experimental procedures, immunofluorescence, and antibodies, as well as metabolomic analysis and databases used please refer to the supporting information (SI).

Abbreviations

- 1,9-DDF
1,9 dideoxyforskolin
- AA
amino acids
- AAT
aspartate amino transferase
- ADP
adenosine diphosphate
- ALT
alanine transaminase
- ATP
adenosine triphosphate
- BCAA

branched chain amino acid

- COX

cytochrome oxidase

- GAC

glutaminase C

- GCs

ganglion cells

- GLUT

glucose transporter

- GS

glutamine synthase

- GTP

guanosine triphosphate

- GABA

gamma amino butyric acid

- GPC

sn-glycero-3-phosphate

- FCCP

carbonyl cyanide-*p*-trifluoromethoxyphenylhydrazone

- GSSG

glutathione disulfide

- INL

inner nuclear layer

- MGC

Müller glial cells

- NAA

N-acetylaspartate

nicotinamide adenine dinucleotide

- ONL

outer nuclear layer

- O-PE

O-phosphoethanolamine

- OXPHOS

oxidative phosphorylation

- PARP

poly(ADP)ribose polymerase

- PCK

phosphoenolpyruvate carboxy kinase

- PKM

pyruvate kinase M

- PNA

peanut agglutinin

- RP

retinitis pigmentosa

- RPE

retinal pigment epithelium

- RPE65

retinal pigment epithelium-specific 65 kDa protein

- SUCLG1

succinate-CoA ligase-1

- TUNEL

terminal UDP nick-end labelling

- UDP

uracil diphosphate

Declarations

Ethics approval and consent to participate

Protocols compliant with the German law on animal protection were reviewed and approved by the institution for animal welfare, veterinary service and laboratory animal science (Einrichtung für Tierschutz, Tierärztlichen Dienst und Labortierkunde) of the University of Tübingen and were following the association for research in vision and ophthalmology (ARVO) statement for the use of animals in vision research.

Availability of data and materials

Raw NMR spectroscopy data is available upon request. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request. This paper does not report original code.

Competing interests

The authors declare no competing financial interests. CT reports a research grant by Bruker BioSpin GmbH.,

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References

1. Country M. W. (2017) **Retinal metabolism: A comparative look at energetics in the retina** *Brain Res* **1672**:50–57
2. Wong-Riley M. T. (2010) **Energy metabolism of the visual system** *Eye Brain* **2**:99–116
3. Swarup A., et al. (2019) **Modulating GLUT1 expression in retinal pigment epithelium decreases glucose levels in the retina: impact on photoreceptors and Müller glial cells** *Am J Physiol Cell Physiol* **316**:C121–c133
4. Warburg O. (1925) **The metabolism of carcinoma cells** *The Journal of Cancer Research* **9**:148–163
5. Okawa H., Sampath A. P., Laughlin S. B., Fain G. L. (2008) **ATP consumption by mammalian rod photoreceptors in darkness and in light** *Curr Biol* **18**:1917–1921
6. Ames A. (1992) **Energy requirements of CNS cells as related to their function and to their vulnerability to ischemia: a commentary based on studies on retina** *Can J Physiol Pharmacol* **70**:S158–164
7. Chinchore Y., Begaj T., Wu D., Drokhlyansky E., Cepko C. L. (2017) **Glycolytic reliance promotes anabolism in photoreceptors** *Elife* **6**
8. Young R. W. (1967) **The renewal of photoreceptor cell outer segments** *J Cell Biol* **33**:61–72
9. Ingram N. T., Fain G. L., Sampath A. P. (2020) **Elevated energy requirement of cone photoreceptors** *Proceedings of the National Academy of Sciences* **117**:19599–19603
10. Sacks D., et al. (2018) **Multisociety Consensus Quality Improvement Revised Consensus Statement for Endovascular Therapy of Acute Ischemic Stroke** *Int J Stroke* **13**:612–632
11. Kanow M. A., et al. (2017) **Biochemical adaptations of the retina and retinal pigment epithelium support a metabolic ecosystem in the vertebrate eye** *Elife* **6**
12. Redmond T. M., et al. (1998) **Rpe65 is necessary for production of 11-cis-vitamin A in the retinal visual cycle** *Nat Genet* **20**:344–351
13. Takata K., Kasahara T., Kasahara M., Ezaki O., Hirano H. (1992) **Ultracytochemical localization of the erythrocyte/HepG2-type glucose transporter (GLUT1) in cells of the blood-retinal barrier in the rat** *Invest Ophthalmol Vis Sci* **33**:377–383
14. Simpson I. A., et al. (2008) **The facilitative glucose transporter GLUT3: 20 years of distinction** *Am J Physiol Endocrinol Metab* **295**:E242–253
15. Rhoads J. M., et al. (1997) **L-glutamine stimulates intestinal cell proliferation and activates mitogen-activated protein kinases** *Am J Physiol* **272**:G943–953
16. Kessler R. J., Tyson C. A., Green D. E. (1976) **Mechanism of uncoupling in mitochondria: uncouplers as ionophores for cycling cations and protons** *Proc Natl Acad Sci U S A* **73**:3141–3145

17. Belhadj S., et al. (2020) **Long-Term, Serum-Free Cultivation of Organotypic Mouse Retina Explants with Intact Retinal Pigment Epithelium** *J Vis Exp* <https://doi.org/10.3791/61868>
18. Joost H. G., Habberfield A. D., Simpson I. A., Laurenza A., Seamon K. B. (1988) **Activation of adenylate cyclase and inhibition of glucose transport in rat adipocytes by forskolin analogues: structural determinants for distinct sites of action** *Mol Pharmacol* **33**:449–453
19. Zhao X., et al. (2018) **Shikonin Inhibits Tumor Growth in Mice by Suppressing Pyruvate Kinase M2-mediated Aerobic Glycolysis** *Scientific Reports* **8**
20. Ortin-Martinez A., et al. (2014) **Number and distribution of mouse retinal cone photoreceptors: differences between an albino (Swiss) and a pigmented (C57/BL6) strain** *PloS one* **9**
21. Jeon C. J., Strettoi E., Masland R. H. (1998) **The major cell populations of the mouse retina** *J Neurosci* **18**:8936–8946
22. Felig P. (1973) **The glucose-alanine cycle** *Metabolism* **22**:179–207
23. Bai P. (2015) **Biology of Poly(ADP-Ribose) Polymerases: The Factotums of Cell Maintenance** *Mol Cell* **58**:947–958
24. Paquet-Durand F., et al. (2007) **Excessive activation of poly(ADP-ribose) polymerase contributes to inherited photoreceptor degeneration in the retinal degeneration 1 mouse** *J Neurosci* **27**:10311–10319
25. Riepe R. E., Norenburg M. D. (1977) **Müller cell localisation of glutamine synthetase in rat retina** *Nature* **268**:654–655
26. Giarmarco M. M., et al. (2020) **Daily mitochondrial dynamics in cone photoreceptors** *Proc Natl Acad Sci U S A* **117**:28816–28827
27. Zabel B. A., et al. (1999) **Human G protein-coupled receptor GPR-9-6/CC chemokine receptor 9 is selectively expressed on intestinal homing T lymphocytes, mucosal lymphocytes, and thymocytes and is required for thymus-expressed chemokine-mediated chemotaxis** *The Journal of experimental medicine* **190**:1241–1256
28. Yudkoff M., Nelson D., Daikhin Y., Erecinska M. (1994) **Tricarboxylic acid cycle in rat brain synaptosomes. Fluxes and interactions with aspartate aminotransferase and malate/aspartate shuttle** *J Biol Chem* **269**:27414–27420
29. Nathwani R. A., Pais S., Reynolds T. B., Kaplowitz N. (2005) **Serum alanine aminotransferase in skeletal muscle diseases** *Hepatology* **41**:380–382
30. Kim W. R., Flamm S. L., Di Bisceglie A. M., Bodenheimer H. C. (2008) **Serum activity of alanine aminotransferase (ALT) as an indicator of health and disease** *Hepatology* **47**:1363–1370
31. Sinha T., Naash M. I., Al-Ubaidi M. R. (2020) **The Symbiotic Relationship between the Neural Retina and Retinal Pigment Epithelium Is Supported by Utilizing Differential Metabolic Pathways** *iScience* **23**
32. Cori C. F., Cori G. T. (1946) **Carbohydrate metabolism** *Annu Rev Biochem* **15**:193–218

33. Rajala R. V. S. (2020) **Aerobic Glycolysis in the Retina: Functional Roles of Pyruvate Kinase Isoforms** *Frontiers in Cell and Developmental Biology* **8**
34. Rueda E. M., et al. (2016) **The cellular and compartmental profile of mouse retinal glycolysis, tricarboxylic acid cycle, oxidative phosphorylation, and ~P transferring kinases** *Molecular vision* **22**:847–885
35. O'Leary F., Campbell M. (2021) **The blood-retina barrier in health and disease** *Febs j* <https://doi.org/10.1111/febs.16330>
36. Diaz-Ruiz R., Rigoulet M., Devin A. (2011) **The Warburg and Crabtree effects: On the origin of cancer cell energy metabolism and of yeast glucose repression** *Biochimica et Biophysica Acta (BBA) - Bioenergetics* **1807**:568–576
37. Crabtree H. G. (1929) **Observations on the carbohydrate metabolism of tumours** *Biochem J* **23**:536–545
38. Balcke G. U., et al. (2011) **Linking energy metabolism to dysfunctions in mitochondrial respiration--a metabolomics in vitro approach** *Toxicol Lett* **203**:200–209
39. Dugan L. L., et al. (1995) **Mitochondrial production of reactive oxygen species in cortical neurons following exposure to N-methyl-D-aspartate** *J Neurosci* **15**:6377–6388
40. Schreiber V., Dantzer F., Ame J. C., de Murcia G. (2006) **Poly(ADP-ribose): novel functions for an old molecule** *Nat Rev Mol Cell Biol* **7**:517–528
41. Venkatesh A., et al. (2015) **Activated mTORC1 promotes long-term cone survival in retinitis pigmentosa mice** *J Clin Invest* **125**:1446–1458
42. Ait-Ali N., et al. (2015) **Rod-derived cone viability factor promotes cone survival by stimulating aerobic glycolysis** *Cell* **161**:817–832
43. Yang M., et al. (2022) **Expression of glucose transporter-2 in murine retina: Evidence for glucose transport from horizontal cells to photoreceptor synapses** *J Neurochem* **160**:283–296
44. Tsacopoulos M., Veuthey A. L., Saravolos S. G., Perrottet P., Tsoupras G. (1994) **Glial cells transform glucose to alanine, which fuels the neurons in the honeybee retina** *J Neurosci* **14**:1339–1351
45. LaNoue K. F., et al. (2001) **Role of specific aminotransferases in de novo glutamate synthesis and redox shuttling in the retina** *J Neurosci Res* **66**:914–922
46. Veeravalli S., et al. (2020) **Flavin-Containing Monooxygenase 1 Catalyzes the Production of Taurine from Hypotaurine** *Drug Metab Dispos* **48**:378–385
47. Sumizu K. (1962) **Oxidation of hypotaurine in rat liver** *Biochim Biophys Acta* **63**:210–212
48. Zheng J. (2012) **Energy metabolism of cancer: Glycolysis versus oxidative phosphorylation (Review)** *Oncol Lett* **4**:1151–1157
49. Pfeiffer T., Schuster S., Bonhoeffer S. (2001) **Cooperation and competition in the evolution of ATP-producing pathways** *Science* **292**:504–507

50. Lai J. C., Behar K. L. (1993) **Glycolysis-citric acid cycle interrelation: a new approach and some insights in cellular and subcellular compartmentation** *Dev Neurosci* **15**:181–193
51. al J. Du et (2016) **Phototransduction Influences Metabolic Flux and Nucleotide Metabolism in Mouse Retina** *J Biol Chem* **291**:4698–4710
52. Gebhard R. (1991) **Cytochemical demonstration of aspartate aminotransferase activity in the rat retina** *Brain Res* **539**:337–341
53. Moffett J. R., Ross B., Arun P., Madhavarao C. N., Namboodiri A. M. A. (2007) **N-Acetylaspartate in the CNS: from neurodiagnostics to neurobiology** *Progress in neurobiology* **81**:89–131
54. Dadsetan S., et al. (2013) **Brain alanine formation as an ammonia-scavenging pathway during hyperammonemia: effects of glutamine synthetase inhibition in rats and astrocyte-neuron co-cultures** *J Cereb Blood Flow Metab* **33**:1235–1241
55. Igarashi H., Suzuki Y., Huber V. J., Ida M., Nakada T. (2015) **N-acetylaspartate decrease in acute stage of ischemic stroke: a perspective from experimental and clinical studies** *Magn Reson Med Sci* **14**:13–24
56. Yan H. D., Ishihara K., Serikawa T., Sasa M. (2003) **Activation by N-acetyl-L-aspartate of acutely dissociated hippocampal neurons in rats via metabotropic glutamate receptors** *Epilepsia* **44**:1153–1159
57. Moffett J., Arun P., Ariyannur P., Namboodiri A. (2013) **N-Acetylaspartate reductions in brain injury: impact on post-injury neuroenergetics, lipid synthesis, and protein acetylation** *Frontiers in Neuroenergetics* **5**
58. Alba-Arbalat S., et al. (2021) **In Vivo Molecular Changes in the Retina of Patients With Multiple Sclerosis** *Investigative ophthalmology & visual science* **62**:11–11

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Reviewer #1 (Public Review):

Summary: In the manuscript by Chen et al. entitled, "The retina uncouples glycolysis and oxidative phosphorylation via Cori-, Cahill-, and mini-Krebs-cycle", the authors look to provide insight on retinal metabolism and substrate utilization by using a murine explant model with various pharmacological treatments in conjunction with metabolomics. The authors conclude that photoreceptors, a specific cell within the explant, which also includes retinal pigment epithelium (RPE) and many other types of cells, are able to uncouple glycolytic and Krebs-cycle metabolism via three different pathways: 1) the mini-Krebs-cycle, fueled by glutamine and branched-chain amino acids; 2) the alanine-generating Cahill-cycle; and 3) the lactate-releasing Cori-cycle. While intriguing if determined to be true, these cell-specific conclusions are called into question due to the ex vivo experimental setup with the inclusion of RPE, the fact that the treatments were not cell-specific nor targeted at an enzyme specific to a certain cell within the retina, and no stable isotope tracing nor mitochondrial function assays were performed. Hence, without significant cell-specific methods and future experimentation, the primary claims are not supported.

Strengths: This study attempts to improve on the issues that have limited the results obtained from previous ex vivo retinal explant studies by culturing in the presence of the RPE, which is a major player in the outer retinal metabolic microenvironment. Additionally, the study utilizes multiple pharmacologic methods to define retinal metabolism and substrate utilization.

Weaknesses: A major weakness of this study is the lack of in vivo supporting data. Explant cultures remove the retina from its dual blood supply. Typically, retinal explant cultures are done without RPE. However, the authors included RPE in the majority of experimental conditions herein. However, it is unclear if the metabolomics samples included the RPE or not. The inclusion of the RPE, which is metabolically active and can be altered by the

treatments investigated herein, further confounds the claims made regarding the neuroretina. Considering the pharmacologic treatments utilized with the explant cultures are not cell-specific and/or have significant off-target effects, it is difficult to ascertain that the metabolic changes are secondary to the effects on photoreceptors alone, which the authors claim. Additionally, the explants are taken at a very early age when photoreceptors are known to still be maturing. No mention or data is presented on how these metabolic changes are altered in retinal explants after photoreceptors have fully matured. Likewise, significant assumptions are made based on a single metabolomics experiment with no stable isotope tracing to support the pathways suggested. While the authors use immunofluorescence to support their claims at multiple points, demonstrating the presence of certain enzymes in the photoreceptors, many of these enzymes are present throughout the retina and likely the RPE. Finally, the claims presented here are in direction contradiction to recent *in vivo* studies that used cell-specific methods when examining retinal metabolism. No discussion of this difference in results is attempted.

Reviewer #2 (Public Review):

Summary: The authors aim to learn about retinal cell-specific metabolic pathways, which could substantially improve the way retinal diseases are understood and treated. They culture *ex vivo* mouse retinas for 6 days with 2 - 4 days of various drug treatments targeting different metabolic pathways or by removing the RPE/choroid tissue from the neural retina. They then look at photoreceptor survival, stain for various metabolic enzymes, and quantify a broad panel of metabolites. While this is an important question to address, the results are not sufficient to support the conclusions.

Strengths: The questions the authors are exploring are extremely valuable and I commend the authors and working to learn more about retina metabolism. The different sensitivity of the cones to various drugs is interesting and may suggest key differences between rods and cones. The authors also provide a thoughtful discussion of various metabolic pathways in the context of previous publications.

Weaknesses: As the authors point out, *ex vivo* culture models allow for control over multiple aspects of the environment (such as drug delivery) not available *in vivo*. *Ex vivo* cultures can provide good hints as to what pathways are available between interacting tissues. However, there are many limitations to *ex vivo* cultures, including shifting to a very artificial culture media condition that is extremely different than the native environment of the retina. It is well appreciated that cells have flexible metabolism and will adapt to the conditions provided. Therefore, observations of metabolic responses obtained under culture conditions need to be interpreted with caution, they indicate what the tissue is doing under those specific conditions (which include cells adapting and dying).

Chen et al use pharmacological interventions to the impact of various metabolic pathways on photoreceptor survival and "long term" metabolic changes. The dose and timing of these drug treatments are not examined though. It is also hard to know how these drugs penetrate the tissue and it needs to be validated that the intended targets are being accurately hit. These relatively long-term treatments should be causing numerous downstream changes to metabolism, cell function, and survival, which makes looking at a snapshot of metabolite levels hard to interpret. It would be more valuable to look at multiple time points after drug treatment, especially early time points (closer to 1 hr). The authors use metabolite ratios to make conclusions about pathway activity. It would be more valuable to directly measure pathway activity by looking at metabolite production rates in the media and/or with metabolic tracers again in time scales closer to minutes and hours instead of days.

It is not clear from the text if the *ex vivo* samples with RPE/choroid intact are analyzed for metabolomics with the RPE/choroid still intact or if this is removed. If it is not removed, the comparison to the retina without RPE/choroid needs to be re-interpreted for the contribution

of metabolites from the added tissue. The composition of the tissue is different and cannot be disentangled from the changes to the neural retina specifically.

While the data is interesting and may give insights into some rod and cone-specific metabolic susceptibility, more work is needed to validate these conclusions. Given the limitations of the model the authors have over-interpreted their findings and the conclusions are not supported by the results. They need to either dramatically limit the scope of their conclusions or validate these hypotheses with additional models and tools.

Reviewer #3 (Public Review):

Summary:

The neural retina is one of the most energetically active tissues in the body and research into retinal metabolism has a rich history. Prevailing dogma in the field is that the photoreceptors of the neural retina (rods and cones) are heavily reliant on glycolysis, and as oxygen tension at the level of photoreceptors is very low, these specialized sensory neurons carry out aerobic glycolysis, akin to the Warburg effect in cancer cells. It has been found that this unique metabolism changes in many retinal diseases, and targeting retinal metabolism may be a viable treatment strategy. The neural retina is composed of 11 different cell types, and many research groups over the past century have contributed to our current understanding of cell-specific metabolism of retinal cells. More recently, it has been shown in mouse models and co-culture of the mouse neural retina with human RPE cultures that photoreceptors are reliant on the underlying retinal pigment epithelium for supplying nutrients. Chen and colleagues add to this body of work by studying an ex vivo culture of the developing mouse retina that maintained contact with the retinal pigment epithelium. They exposed such ex vivo cultures to small molecule inhibitors of specific metabolic pathways, performing targeted metabolomics on the tissue and staining the tissue with key metabolic enzymes to lay the groundwork for what metabolic pathways may be active in particular cell types of the retina. The authors conclude that rod and cone photoreceptors are reliant on different metabolic pathways to maintain their cell viability - in particular, that rods rely on oxidative phosphorylation and cones rely on glycolysis. Further, their data support multiple mechanisms whereby glycolysis may occur simultaneously with anapleurosis to provide abundant energy to photoreceptors. The data from metabolomics revealed several novel findings in retinal metabolism, including the use of glutamine to fuel the mini-Krebs cycle, the utilization of the Cahill cycle in photoreceptors, and a taurine/hypotaurine shuttle between the underlying retinal pigment epithelium and photoreceptors to transfer reducing equivalents from the RPE to photoreceptors. In addition, this study provides robust quantitative metabolomics datasets that can be compared across experiments and groups. The use of this platform will allow for rapid testing of novel hypotheses regarding the metabolic ecosystem in the neural retina.

Strengths:

The data on differences in the susceptibility of rods and cones to mitochondrial dysfunction versus glycolysis provides novel hypothesis-generating conjectures that can be tested in animal models. The multiple mechanisms that allow anapleurosis and glycolysis to run side-by-side add significant novelty to the field of retinal metabolism, setting the stage for further testing of these hypotheses as well.

Weaknesses:

Almost all of the conclusions from the paper are preliminary, based on data showing enzymes necessary for a metabolic process are present and the metabolites for that process are also present. However, to truly prove whether these processes are happening, C13 labeling or knock-out or over-expression experiments are necessary. Further, while there is good data that RPE cultures in vitro strongly recapitulate RPE phenotypes in vivo, ex vivo neural retina cultures undergo rapid death. Thus, conclusions about metabolism from

explants should either be well correlated with existing literature or lead to targeted in vivo studies. This paper currently lacks both.